

- **Data Needed.** This section describes the block-level and county-level data underlying the forecasts.
- **PopGrid.** This section reviews the PopGrid software application, which aggregates block-level population data to whatever grid definition might be need.

J.1.1 How BenMAP Forecasts Population

In calculating the population in age groups that may include a portion of one of the pre-specified demographic groups in Table J-1, BenMAP assumes the population is uniformly distributed in the age group. For example, to calculate the number of children ages 3 through 12, BenMAP calculates:

$$age_{3-12} = \frac{1}{2} \times age_{1-4} + age_{5-9} + \frac{3}{5} \times age_{10-14}$$

To estimate population levels for the years after the last Census in 2020, BenMAP scales the 2020 Census-based estimate with the ratio of the county-level forecast for the future year of interest over the 2020 county-level population level. Woods & Poole (2015) provides the county-level population forecasts until 2055 used to calculate the scaling ratios; these data are discussed in detail below.

In the simplest case, where one is forecasting a single population variable, say, children ages 4 to 9 in the year 2030, BenMAP calculates:

$$age_{4-9,g,2030} = age_{4-9,g,2020} \times \frac{age_{4-9,county,2030}}{age_{4-9,county,2020}}$$

Where the gth population grid-cell is wholly located within a given county.

In the case, where the gth grid-cell includes “n” counties in its boundary, the situation is somewhat more complicated. BenMAP first estimates the fraction of individuals in a given age group (e.g., ages 4 to 9) that reside in the part of each county within the gth grid-cell. BenMAP calculates this fraction by simply dividing the population all ages of a given county within the gth grid-cell by the total population in the gth grid-cell:

$$fraction\ of\ age_{4-9,g\ in\ county_c} = \frac{age_{all,g\ in\ county_c}}{age_{all,g}}$$

Multiplying this fraction with the number of individuals ages 4 to 9 in the year 2020 gives an estimate of the number of individuals ages 4 to 9 that reside in the fraction of the county within the gth grid-cell in the year 2020:

$$age_{4-9,g\ in\ county_c,2020} = age_{4-9,g,2020} \times fraction\ age_{4-9,g\ in\ county_c}$$

To then forecast the population in 2030, we scale the 2020 estimate with the ratio of the county projection for 2030 to the county projection for 2020:

$$age_{4-9,g \text{ in county } c,2030} = age_{4-9,g \text{ in county } c,2020} \times \frac{age_{4-9, \text{county } c,2030}}{age_{4-9, \text{county } c,2020}}$$

Combining all these steps for “n” counties within the gth grid-cell, we forecast the population of persons ages 4 to 9 in the year 2030 as follows:

$$age_{4-9,g,2030} = \sum_{c=1}^n age_{4-9,g,2020} \times \frac{total \ pop_g \ \text{in county } c}{total \ pop_g} \times \frac{age_{4-9, \text{county } c,2030}}{age_{4-9, \text{county } c,2020}}$$

In the case where there are multiple age groups and multiple counties, BenMAP first calculates the forecasted population level for individual age groups, and then combines the forecasted age groups. In calculating the number of children ages 4 to 12, BenMAP calculates:

$$age_{4-9,g,2030} = \sum_{c=1}^n age_{4-9,g,2020} \times \frac{total \ pop_g \ \text{in county } c}{total \ pop_g} \times \frac{age_{4-9, \text{county } c,2030}}{age_{4-9, \text{county } c,2020}}$$

$$age_{10-14,g,2030} = \sum_{c=1}^n age_{10-14,g,2020} \times \frac{total \ pop_g \ \text{in county } c}{total \ pop_g} \times \frac{age_{10-14, \text{county } c,2030}}{age_{10-14, \text{county } c,2020}}$$

$$age_{4-12,g,2030} = age_{4-9,g,2030} + \frac{3}{5} \times age_{10-14,g,2030}$$

To estimate population for 2055, we extrapolate Woods and Poole projections from the 2045 to 2050 period:

$$W_{2055,i} = W_{2050,i} * \frac{W_{2050,i}}{W_{2045,i}}$$

where W is the growth weight (relative to 2020) and i is each demographic cell (i.e., unique combinations of county, gender, ethnicity, race, and age range).

J.1.2 Data Needed for Forecasting

Underlying the population forecasts in BenMAP there are block-level databases used to provide year 2020 population estimates and a county-level database of forecast ratios. Both files have the same set of 304 race-ethnicity-gender-age population groups.

The block-level data is typically not used directly in BenMAP, and instead is used with the PopGrid software (described below) to provide year 2020 estimates for a grid definition of interest (e.g., 12 kilometer CMAQ grid). The output from PopGrid with the year 2020 population estimates can then be loaded into BenMAP.

The county-level data comes pre-installed in the U.S. setup, and is not something that the user needs to load. These data are simply county-level ratios of a year (2000-2019, 2021-2050) and year 2020 population data for each county and each of the 304 race-ethnicity-gender-age population groups.

We describe the development of each database below.

J.1.2.1 Block-Level Census 2020

There are about five million “blocks” in the United States, and for each block we have 304 race-ethnicity-gender-age groups. The block-level population database is created separately for each state, in order to make the data more manageable. (A single national file of block data would be about six gigabytes.)

The initial block file from the U.S. Census Bureau is not in the form needed. The block data has 7 racial categories and 23 age groups, as opposed to the 4 and 19 used in BenMAP. Table J-2 summarizes the initial set of variables and the final desired set of variables.

Table J-2. Race, Ethnicity and Age Variables in 2020 Census Block Data

Type	Race	Ethnicity	Gender	Age
Initial Variables (DHC file)	White Alone, Black Alone, Native American Alone, Asian Alone, Pacific Islander/Hawaiian Alone, Other Alone, Two or More Alone	--	Male, Female	0-4, 5-9, 10-14, 15-17, 18-19, 20, 21, 22-24, 25-29, 30-34, 35-39, 40-44, 45-49, 50-54, 55-59, 60-61, 62-64, 65-66, 67-69, 70-74, 75-79, 80-84, 85+
Final Desired Variables	White, African-American, Asian-American, Native-American	Hispanic, Non-Hispanic	Female, Male	<1, 1-4, 5-9, 10-14, 15-19, 20-24, 25-29, 30-34, 35-39, 40-44, 45-49, 50-54, 55-59, 60-64, 65-69, 70-74, 75-79, 80-84, 85+

The initial set of input files are as follows.

- Census 2020 block-level and tract-level files (Demographic and Housing Characteristics (DHC))
 Data: <https://www.census.gov/data/tables/2023/dec/2020-census-dhc.html>
 Docs: <https://www2.census.gov/programs-surveys/decennial/2020/technical-documentation/complete-tech-docs/demographic-and-housing-characteristics-file-and-demographic-profile/2020census-demographic-and-housing-characteristics-file-and-demographic-profile-techdoc.pdf>
- Census 2010 MARS national-level summary
 Docs: <https://www2.census.gov/programs-surveys/popest/technical-documentation/methodology/modified-race-summary-file-method/mrsf2010.pdf>

The DHC tract-level and MARS data, as described below, are needed to reorganize the variables that come initially in the block-level DHC file. (For the sake of completeness, we note that there exists a county-level Census 2010 MARS file; however, due to major population count discrepancies between the county-level MARS file and block-level DHC file, we used only the nation-level summary table. Tables in MARS documentation file did not have the discrepancies that the county-level file had. We were unable to get an adequate explanation of this from the U. S. Census.)

The steps in preparing the data are as follows:

1. Adjust Age-classifications:

We combined some age groups in the block-level DHC data to match the age groups wanted for BenMAP. For example, we combined age groups 15-17 and 18-19 to create the 15-19 age group used in BenMAP. Then, in the case of the 0-4 age group, we split it into <1 and 1-4 using the tract-level DHC data, which gave us the fraction of 0-4 year-olds who are <1.

2. Fill in Missing Racial-Ethnic Interactions:

We used the tract-level DHC data to calculate the fraction of Hispanics in each ethnically-aggregated subpopulation from the block-level data, by age and sex. We used these fractions to distribute each age-sex-race-block-level datum into Hispanics and non-Hispanics.

3. Assign “Other” and “Multi-Racial” to the Remaining Four Racial Categories:

We assign the “Other” race category in two steps. First, based on the national MARS data, we estimated how many people in the “multi-racial” category checked off “some other race” as one of their races, for Hispanics and non-Hispanics separately. In each age-sex-race-block-level datum, we added those people to “other race” category to create the re-distribution pool, analogously to the method implemented by Census while creating MARS data (see U.S. Census Bureau, 2012, Table J-3, below). Second, based on the national re-allocation fractions for Hispanics and non-Hispanics (derived from the MARS data), we assigned the “Other” race into the four races of interest and “multi-race”.

After the assignment of the “Other” race category, we then assigned “multi-racial” category to the four racial categories, using state fractions of these races in each age-sex-race-block-level datum.